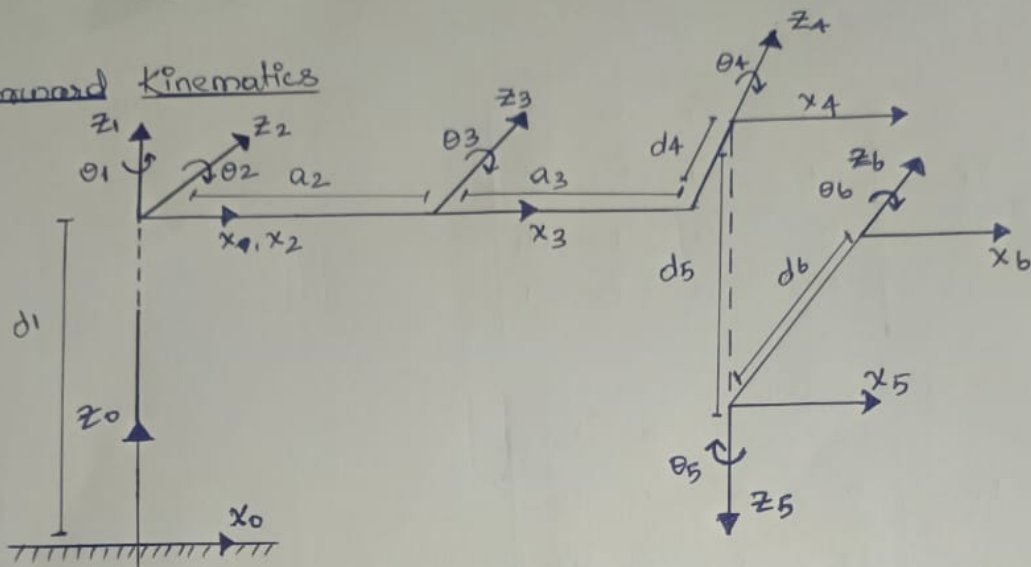


UR5E Kinematics and Dynamics

①

* Forward Kinematics



Modified (Craig's) convention

Modified D-H Table.

	i	α_{i-1}	a_{i-1}	d_i	θ_i
1	1	0	0	d_1	θ_1
2	2	$\frac{3\pi}{2}$	0	0	θ_2
3	3	0	a_2	0	θ_3
4	4	0	a_3	d_4	θ_3
5	5	$3\pi/2$	0	d_5	θ_5
6	6	$\pi/2$	0	d_6	θ_6

WKT:

$${}^{i-1}T_i = \begin{bmatrix} \cos \theta_i & -\sin \theta_i & 0 & a_{i-1} \\ \sin \theta_i \cos \alpha_{i-1} & \cos \theta_i \cos \alpha_{i-1} & -\sin \alpha_{i-1} & -d_i \sin \alpha_{i-1} \\ \sin \theta_i \sin \alpha_{i-1} & \cos \theta_i \sin \alpha_{i-1} & \cos \alpha_{i-1} & d_i \cos \alpha_{i-1} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Now deriving each transformation matrices using DH-table.

for $i=1$,

$${}^0T_1 = \begin{vmatrix} c_1 & -s_1 & 0 & 0 \\ s_1 & c_1 & 0 & 0 \\ 0 & 0 & 1 & d_1 \\ 0 & 0 & 0 & 1 \end{vmatrix}$$

for $i=2$,

$${}^2T_1 = \begin{vmatrix} c_2 & -s_2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -s_2 & -c_2 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix}$$

for $i=3$,

$${}^3T_2 = \begin{vmatrix} c_3 & -s_3 & 0 & a_2 \\ s_3 & c_3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix}$$

for $i=4$,

$${}^4T_3 = \begin{vmatrix} c_4 & -s_4 & 0 & a_3 \\ s_4 & c_4 & 0 & 0 \\ 0 & 0 & 1 & d_4 \\ 0 & 0 & 0 & 1 \end{vmatrix}$$

for $i=5$,

$${}^4T_5 = \begin{vmatrix} c_5 & -s_5 & 0 & 0 \\ 0 & 0 & 1 & d_5 \\ -s_5 & -c_5 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix}$$

for $i=6$,

$${}^5T_6 = \begin{vmatrix} c_6 & -s_6 & 0 & 0 \\ 0 & 0 & -1 & -d_6 \\ s_6 & c_6 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix}$$

Now we have a forward kinematic model, to determine the end-effector's pose w.r.t. to the base.

$$\begin{pmatrix} {}^0T_6 \\ {}^0T_E \end{pmatrix} = {}^0T_1 T_2 T_3 T_4 T_5 T_6$$

* Differential Inverse Kinematics :

$$\text{WKT: } (\text{Twist})_{\text{End-Effector}} = J(q) \dot{q} = V_E$$

$$\rightarrow (V_E = J(q) \dot{q})$$

$$\rightarrow (\dot{q} = J^{-1}(q) V_E)$$

Thus we need to find Jacobian J_0^6 ; so we can find J^+ to drive the robot using Diff. IK.

* Finding Jacobian using Velocity propagation Method:

WKT: Angular Velocity :

$${}^i \omega_i = {}^i_{i-1} R \cdot {}^{i-1} \omega_{i-1} + \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_i \end{bmatrix}$$

Linear Velocity :

$${}^i V_i = {}^i_{i-1} R \left({}^{i-1} V_{i-1} + {}^{i-1} \omega_{i-1} \times {}^{i-1} P_i \right)$$

Now:

For Link 1:

$${}^1 \omega_1 = \begin{bmatrix} c_1 & s_1 & 0 \\ -s_1 & c_1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{bmatrix} \quad \text{--- (1w)}$$

$${}^1 V_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \text{ because } {}^0 V_0 = 0_{3 \times 1} = {}^0 \omega_0$$

--- (2w)

For Link 2:

$${}^2 \omega_2 = \begin{bmatrix} c_2 & 0 & -s_2 \\ -s_2 & 0 & -c_2 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_2 \end{bmatrix}$$

$${}^2 \omega_2 = \begin{bmatrix} -s_2 \dot{\theta}_1 \\ -c_2 \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix} \quad \text{--- (2w)}$$

$${}^2 V_2 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \text{ because } {}^1 V_1 = {}^1 P = 0_{3 \times 1}$$

--- (2v)

(3)

For Link 3:

$${}^3\omega_3 = \begin{bmatrix} c_3 & s_3 & 0 \\ -s_3 & c_3 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -s_2 \dot{\theta}_1 \\ -c_2 \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_3 \end{bmatrix}$$

$${}^3\omega_3 = \begin{bmatrix} -c_3 s_2 \dot{\theta}_1 - s_3 c_2 \dot{\theta}_1 \\ + s_3 s_2 \dot{\theta}_1 - c_2 c_3 \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_3 \end{bmatrix}$$

$$(R_1) = -\dot{\theta}_1 (s_2 c_3 + c_2 s_3) = -\dot{\theta}_1 (s_{23})$$

$$\circ^\circ (s_a c_b + s_b c_a = s_{ab} = s(a+b))$$

$$(R_2) = -\dot{\theta}_1 (c_2 c_3 - s_2 s_3) = -\dot{\theta}_1 (c_{23})$$

$$\circ^\circ (c_a c_b - s_a s_b = c_{ab} = c(a+b))$$

$$\Rightarrow {}^3\omega_3 = \begin{bmatrix} -\dot{\theta}_1 s_{23} \\ -\dot{\theta}_1 c_{23} \\ \dot{\theta}_{23} \end{bmatrix} \quad \text{--- (3w)}$$

$${}^3V_3 = {}^3R ({}^2V_2 + \underbrace{{}^2\omega_2 \times {}^2P_3}) \text{ now.}$$

$$\begin{vmatrix} i & j & k \\ -s_2 \dot{\theta}_1 & -c_2 \dot{\theta}_1 & \dot{\theta}_2 \\ a & 0 & 0 \end{vmatrix}$$

$${}^2\omega_2 \times {}^2P_3 = \begin{bmatrix} 0 \\ a \dot{\theta}_2 \\ a_2 c_2 \dot{\theta}_1 \end{bmatrix}$$

$${}^3V_3 = \begin{bmatrix} c_3 & s_3 & 0 \\ -s_3 & c_3 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ a_2 \ddot{\theta}_2 \\ a_2 c_2 \dot{\theta}_1 \end{bmatrix}$$

$${}^3V_3 = \begin{bmatrix} s_3 a_2 \ddot{\theta}_2 \\ c_3 a_2 \ddot{\theta}_2 \\ a_2 c_2 \dot{\theta}_1 \end{bmatrix} \quad \text{--- } (3V)$$

For Link 4 :

$${}^4\omega_4 = \begin{bmatrix} c_4 & s_4 & 0 \\ -s_4 & c_4 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -s_{23} \dot{\theta}_1 \\ -c_{23} \dot{\theta}_1 \\ \dot{\theta}_{23} \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_4 \end{bmatrix}$$

$${}^4\omega_4 = \begin{bmatrix} -s_{23} c_4 \dot{\theta}_1 - c_{23} s_4 \dot{\theta}_1 \\ s_{23} s_4 \dot{\theta}_1 - c_{23} c_4 \dot{\theta}_1 \\ \dot{\theta}_{23} \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_4 \end{bmatrix}$$

$$\text{III}^{\text{ly}}. \quad {}^4\omega_4 = \begin{bmatrix} -s_{234} \dot{\theta}_1 \\ -c_{234} \dot{\theta}_1 \\ \dot{\theta}_{234} \end{bmatrix} \quad \text{--- } (4\omega)$$

$${}^4V_4 = {}^3R_4 \left({}^3V_3 + \underbrace{{}^3\omega_3 \times {}^3P_4} \right)$$

$$\begin{vmatrix} i & j & k \\ -s_{23} \dot{\theta}_1 & -c_{23} \dot{\theta}_1 & \dot{\theta}_{23} \\ a_3 & 0 & d_4 \end{vmatrix}$$

$${}^3\omega_3 \times {}^3P_4 = \begin{bmatrix} -d_4 c_{23} \dot{\theta}_1 \\ a_3 \dot{\theta}_{23} + d_4 s_{23} \dot{\theta}_1 \\ a_3 c_{23} \dot{\theta}_1 \end{bmatrix}$$

$$\underbrace{{}^3V_3 + {}^3\omega_3 \times {}^3_4P}_{A.} = \begin{bmatrix} s_3 a_2 \ddot{\theta}_2 - d_4 c_{23} \ddot{\theta}_1 \\ c_3 a_2 \ddot{\theta}_2 + a_3 \ddot{\theta}_{23} + d_4 s_{23} \ddot{\theta}_1 \\ a_2 c_2 \ddot{\theta}_1 + a_2 c_{23} \ddot{\theta}_1 \end{bmatrix} \quad (1)$$

$$\text{Now } {}^4_3R(A) = \begin{bmatrix} c_4 & s_4 & 0 \\ -s_4 & c_4 & 0 \\ 0 & 0 & 1 \end{bmatrix} \downarrow [A]_{3 \times 1}$$

$${}^4_3R(A) = \begin{bmatrix} c_4 (s_3 a_2 \ddot{\theta}_2 - d_4 c_{23} \ddot{\theta}_1) + s_4 (c_3 a_2 \ddot{\theta}_2 + a_3 \ddot{\theta}_{23} + d_4 s_{23} \ddot{\theta}_1) \\ -s_4 (s_3 a_2 \ddot{\theta}_2 - d_4 c_{23} \ddot{\theta}_1) + c_4 (c_3 a_2 \ddot{\theta}_2 + a_3 \ddot{\theta}_{23} + d_4 s_{23} \ddot{\theta}_1) \\ (a_2 c_2 + a_2 c_{23}) \ddot{\theta}_1 \end{bmatrix}$$

$$\begin{matrix} 4 \\ 3 \text{ (or)} \\ 4 V_4 \end{matrix} R(A) = \begin{bmatrix} a_2 s_{23} \ddot{\theta}_2 + a_3 s_4 \ddot{\theta}_{23} - d c_{234} \ddot{\theta}_1 \\ a_2 c_{34} \ddot{\theta}_2 + a_3 c_4 \ddot{\theta}_{23} + d s_{234} \ddot{\theta}_1 \\ (a_2 c_2 + a_2 c_{23}) \ddot{\theta}_1 \end{bmatrix} - (4V)$$

For Link 5:

$$S_{\omega_5} = \begin{bmatrix} c_5 & 0 & -s_5 \\ -s_5 & 0 & -c_5 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} -s_{234} \ddot{\theta}_1 \\ -c_{234} \ddot{\theta}_1 \\ \ddot{\theta}_{234} \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \ddot{\theta}_5 \end{bmatrix}$$

$$S_{\omega_5} = \begin{bmatrix} -(c_5 s_{234} \ddot{\theta}_1 + s_5 \ddot{\theta}_{234}) \\ s_5 s_{234} \ddot{\theta}_1 - c_5 \ddot{\theta}_{234} \\ \ddot{\theta}_5 - c_{234} \ddot{\theta}_1 \end{bmatrix} \quad (5\omega)$$

(5)

$${}^S V_S = {}^4 R_S \left(\underbrace{{}^4 V + {}^4 \omega \times {}^4 P}_{{\det.} \quad V^*} \right)$$

$$V^* = \begin{bmatrix} {}^4 V + \underbrace{{}^4 \omega \times {}^4 P}_{{\det.} \quad V^*} \end{bmatrix}$$

$$\begin{vmatrix} i & j & k \\ -s_{234}\dot{\theta}_1 & -c_{234}\dot{\theta}_1 & \dot{\theta}_{234} \\ 0 & d_5 & 0 \end{vmatrix} = \begin{bmatrix} -d_5 \dot{\theta}_{234} \\ 0 \\ -d_5 s_{234} \dot{\theta}_1 \end{bmatrix}$$

$$V^* = \begin{bmatrix} a_2 s_{34} \dot{\theta}_2 + a_3 s_4 \dot{\theta}_{23} - d_4 c_{234} \dot{\theta}_1 - d_5 \dot{\theta}_{234} \\ a_2 c_{34} \dot{\theta}_2 + a_3 c_4 \dot{\theta}_{23} + d_4 s_{234} \dot{\theta}_1 \\ (a_2 c_2 + a_3 c_{23} + d_5 s_{234}) \dot{\theta}_1 \end{bmatrix} = \begin{bmatrix} V_x^* \\ V_y^* \\ V_z^* \end{bmatrix}$$

$${}^S V_S = \begin{bmatrix} c_5 & 0 & -s_5 \\ -s_5 & 0 & -c_5 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} V_x^* \\ V_y^* \\ V_z^* \end{bmatrix} = \begin{bmatrix} c_5 V_x^* - s_5 V_z^* \\ -s_5 V_x^* - c_5 V_z^* \\ V_y^* \end{bmatrix}$$

SV

For Link 6:

$${}^6w_6 = \begin{bmatrix} c_6 & 0 & s_6 \\ -s_6 & 0 & c_6 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} -(c_5 s_{234} \ddot{\theta}_1 + s_5 \ddot{\theta}_{234}) \\ s_5 s_{234} \ddot{\theta}_1 - c_5 \ddot{\theta}_{234} \\ \dot{\theta}_5 - c_{234} \dot{\theta}_1 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_6 \end{bmatrix}$$

$${}^6w_6 = \begin{bmatrix} -c_6 (c_5 s_{234} \ddot{\theta}_1 + s_5 \ddot{\theta}_{234}) + s_6 (\dot{\theta}_5 - c_{234} \dot{\theta}_1) \\ -s_6 (s_5 s_{234} \ddot{\theta}_1 - c_5 \ddot{\theta}_{234}) + c_6 (\dot{\theta}_5 - c_{234} \dot{\theta}_1) \\ c_5 \ddot{\theta}_{234} - s_5 s_{234} \ddot{\theta}_1 + \dot{\theta}_6 \end{bmatrix}$$

$${}^6v_6 = {}^S_R \left({}^S_V + \omega_S \times {}^S_P \right)$$

$$\begin{matrix} i & j \\ \swarrow & \searrow \\ B & \end{matrix}$$

$$B = \begin{vmatrix} i & j & k \\ -(c_5 s_{234} \ddot{\theta}_1 + s_5 \ddot{\theta}_{234}) & s_5 s_{234} \ddot{\theta}_1 - c_5 \ddot{\theta}_{234} & \dot{\theta}_5 - c_{234} \dot{\theta}_1 \\ 0 & -d_6 & 0 \end{vmatrix}$$

$$B = \begin{bmatrix} d_6 (\dot{\theta}_5 - c_{234} \dot{\theta}_1) \\ 0 \\ d_6 (c_5 s_{234} \ddot{\theta}_1 + s_5 \ddot{\theta}_{234}) \end{bmatrix}$$

(6)

$$U^* = {}^S V_5 + (\underbrace{{}^S \omega_5 \times}_B {}^S P_6) = \begin{bmatrix} {}^S v_{5,x} + d_6(\dot{\theta}_5 - c_{234}\dot{\theta}_1) \\ {}^S v_{5,y} \\ {}^S v_{5,z} + d_6(c_5 s_{234}\dot{\theta}_1 + s_5\dot{\theta}_{234}) \end{bmatrix} = \begin{bmatrix} U_x^* \\ U_y^* \\ U_z^* \end{bmatrix}$$

$${}^6 V_6 = \begin{bmatrix} c_6 & 0 & s_6 \\ -s_6 & 0 & c_6 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} U_x^* \\ U_y^* \\ U_z^* \end{bmatrix} = \begin{bmatrix} c_6 U_x^* + s_6 U_z^* \\ -s_6 U_x^* + c_6 U_z^* \\ -U_y^* \end{bmatrix} \quad (6V)$$